An Impossibility Theorem for Stable and Equitable Feature Attribution in Sequential Gradient Boosting

1 Formal Setup

Consider a supervised learning problem with P features $\mathcal{P} = \{1, \ldots, P\}$ and response y. The features admit a *correlation partition* $\mathcal{G} = \{G_\ell\}_{\ell=1}^{L}$ such that for every group G_ℓ and every pair $j, k \in G_\ell$, the pairwise Pearson correlation satisfies $\mathrm{Corr}(x_j, x_k) = \rho$ for some $\rho \in (0, 1)$.

Definition 1 (Data-generating process)**.** *The true DGP is*

$$y = \sum_{\ell=1}^{L} \sum_{j \in G_\ell} \beta_j \, x_j + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \sigma^2), \tag{1}$$

where all features within each group share a common coefficient: $\beta_j = \beta_k$ for all $j, k \in G_\ell$. The features $(x_1, \ldots, x_P)$ are jointly Gaussian with unit marginal variances; within each group G_ℓ, all pairs have correlation ρ, and features in distinct groups are uncorrelated.

Definition 2 (Sequential gradient boosting)**.** *A gradient-boosted ensemble f is constructed sequentially:*

$$f = \sum_{t=1}^{T} \eta \, h_t, \tag{2}$$

where $\eta > 0$ is the learning rate and each base learner h_t is a regression tree fit to the pseudo-residuals $r_t = y - \sum_{s<t} h_s(x)$. At each internal node, the tree greedily selects the feature–threshold pair (j^*, c^*) maximizing the squared-error reduction. We write $\omega \in \Omega$ for the random seed controlling data sub-sampling, feature sub-sampling, and tie-breaking during training.

Definition 3 (Attributions and metrics)**.** *Let $\phi_j(f)$ denote the interventional TreeSHAP global feature importance for feature j in model f, defined as the mean absolute SHAP value over an explanation set: $\phi_j(f) = \frac{1}{n} \sum_{i=1}^{n} |\phi_j(f; x^{(i)})|$.*

Definition 4 (Stability and Equity)**.** *Let f_ω denote a model trained with random seed ω.*

- ***Stability.*** *A method is δ-stable if for any two independent draws ω, ω', the Spearman rank correlation between the importance vectors satisfies*

$$\mathbb{E}\big[\mathrm{Spearman}\big(\boldsymbol{\phi}(f_\omega),\ \boldsymbol{\phi}(f_{\omega'})\big)\big] \geq 1 - \delta. \tag{3}$$

- ***Equity.*** *A method is γ-equitable if for every group G_ℓ with $\beta_j = \beta_k$ for all $j, k \in G_\ell$ (and $\phi_k(f_\omega) > 0$ for all $k \in G_\ell$),*

$$\mathbb{E}\left[\frac{\max_{j \in G_\ell} \phi_j(f_\omega)}{\min_{k \in G_\ell} \phi_k(f_\omega)}\right] \leq 1 + \gamma. \tag{4}$$

A method satisfies both desiderata simultaneously if it is δ-stable and γ-equitable for small $\delta, \gamma > 0$.

Remark 5 (Positivity of attributions). The assumption $\phi_k(f_\omega) > 0$ for all $k \in G_\ell$ required by Definition 4 holds for $\beta_j > 0$ with sufficient tree depth ($d \geq 2$) and sample size, since every feature in G_ℓ will appear as a split variable in at least one tree with positive probability.

2 First-Mover Concentration

Lemma 6 (First-mover concentration of splits)**.** *Consider a group $G_\ell = \{j_1, \ldots, j_m\}$ with pairwise correlation ρ and equal true coefficients. In a sequential gradient-boosted ensemble f_ω with T trees and bounded depth $d \geq 2$, suppose the first tree h_1 selects feature $j_1 \in G_\ell$ at its root split (determined by seed ω). Then for any other feature $j_q \in G_\ell$, $q \neq 1$, the expected cumulative split counts satisfy*

$$\mathbb{E}_\omega\big[n_{j_1}\big] - \mathbb{E}_\omega\big[n_{j_q}\big] = \Omega(\rho^2 \cdot T). \tag{5}$$

More precisely, the expected split counts satisfy

$$n_{j_1} \approx \frac{1}{2-\rho^2}\,T, \qquad n_{j_q} \approx \frac{1-\rho^2}{2-\rho^2}\,T, \tag{6}$$

giving the gap $n_{j_1} - n_{j_q} = \dfrac{\rho^2}{2-\rho^2}\,T \geq c\rho^2 T$ for a constant $c > 0$.

Proof. Write $x_{j_q} = \rho\, x_{j_1} + \sqrt{1-\rho^2}\, z_q$ where $z_q \perp x_{j_1}$ (possible since features are jointly Gaussian by Theorem 1).

The first tree h_1 is a function of x_{j_1} alone (by assumption). Because $z_q \perp x_{j_1}$, we have $\mathrm{Cov}(h_1, x_{j_q}) = \rho\,\mathrm{Cov}(h_1, x_{j_1})$. Hence h_1 simultaneously captures the ρ-aligned component of x_{j_q}'s signal, leaving only the orthogonal component $\sqrt{1-\rho^2}\, z_q$ as the unique residual signal available to j_q in all subsequent trees.

Summing gains across all T trees, the total split gain attributable to j_q is bounded by $\Theta((1-\rho^2)\,\beta_{j_q}^2\,\mathrm{Var}(z_q)\cdot T)$, while j_1's total gain is $\Theta(\beta_{j_1}^2\,\mathrm{Var}(x_{j_1})\cdot T)$. Since $\beta_{j_1} = \beta_{j_q}$ and $\mathrm{Var}(x_{j_1}) = \mathrm{Var}(z_q) = 1$ within each group (unit marginal variances by Theorem 1), normalising by the shared group signal gives the expected split counts in (6).

One verifies: $n_{j_1} - n_{j_q} = (1-(1-\rho^2))/(2-\rho^2)\cdot T = \rho^2/(2-\rho^2)\cdot T$. Since $\rho^2/(2-\rho^2) \geq \rho^2/2$ for all $\rho \in (0,1)$, there exists a constant $c = 1/2 > 0$ such that

$$\mathbb{E}[n_{j_1}] - \mathbb{E}[n_{j_q}] \;\geq\; c\cdot\rho^2\cdot T. \tag{7}$$

The bounded-depth assumption ($d \geq 2$) ensures each tree captures only a bounded fraction $\lambda < 1$ of the available signal along any feature, so the residual signal accounting above applies iteratively across trees.

Crucially, the identity of the first-mover j_1 is determined by ω (through sub-sampling and tie-breaking) and is *approximately uniform* over G_ℓ by symmetry of the DGP. □

3 Simplified Attribution Model

Assumption 7 (Uniform-contribution model)**.** *Trees have bounded depth d; each split on feature j contributes $\Theta(\eta)$ to $\phi_j(f)$ in expectation; and features appear at all tree depths at comparable rates. We further assume the calibration condition $\eta\sum_k n_k = \eta\,\Theta(T) = \Theta(1)$, i.e., ηT is bounded away from zero and infinity.*

Remark 8. Assumption 7 is a simplification: in general, TreeSHAP weights splits by depth, prediction magnitude, and Shapley combinatorial coefficients, so a root split contributes far more than a leaf split. The Lemma below is conditional on this assumption and gives the essential scaling behaviour; removing it would require tracking depth-weighted split counts, complicating the algebra without changing the qualitative conclusion.

Lemma 9 (TreeSHAP inherits split concentration)**.** *Under Theorem 7, for a tree ensemble* $f = \sum_t h_t$*, the interventional TreeSHAP global importance* $\phi_j(f)$ *satisfies*

$$\phi_j(f) = \Theta\left(\frac{n_j}{\sum_{k=1}^{P} n_k}\right). \tag{8}$$

Consequently, if $n_{j_1} - n_{j_q} = \Omega(\rho^2 \cdot T)$*, then*

$$\phi_{j_1}(f) - \phi_{j_q}(f) = \Omega(\rho^2). \tag{9}$$

Moreover, from the split-count formula (6)*,*

$$\phi_{j_q}(f_\omega) = \Theta\left(\frac{1-\rho^2}{2-\rho^2}\right) \xrightarrow{\rho\to 1} 0, \qquad \phi_{j_1}(f_\omega) = \Theta\left(\frac{1}{2-\rho^2}\right) \xrightarrow{\rho\to 1} \Theta(1). \tag{10}$$

Proof. Interventional TreeSHAP computes, for each tree h_t, the Shapley value of feature j with respect to the prediction function defined by that tree's structure. A feature that never appears as a split variable in h_t receives zero attribution from that tree. For a feature that does appear, its attribution in h_t is determined by the magnitude of the prediction difference at the nodes where it splits, weighted by the Shapley combinatorial coefficients.

Summing over all trees, the global importance $\phi_j(f)$ aggregates contributions from exactly those trees containing splits on j. Under Theorem 7, each split on j contributes $\Theta(\eta)$ to $\phi_j(f)$ in expectation. Therefore

$$\phi_j(f) = \Theta(\eta \cdot n_j) = \Theta\left(\frac{n_j}{\sum_k n_k}\right), \tag{11}$$

since $\sum_k n_k = \Theta(T/\eta)$ for trees of bounded depth and the calibration condition $\eta T = \Theta(1)$ of Theorem 7.

The additive gap (9) follows by applying Theorem 6: the split gap $n_{j_1} - n_{j_q} = \Omega(\rho^2 \cdot T)$ implies

$$\phi_{j_1}(f) - \phi_{j_q}(f) = \Theta\left(\eta \cdot \rho^2 \cdot T \cdot \frac{1}{\sum_k n_k}\right) = \Omega(\rho^2), \tag{12}$$

which is bounded away from zero for any fixed $\rho > 0$.

The limits (10) follow directly by substituting the normalised split counts from (6). □

4 Impossibility Theorem

Theorem 10 (Impossibility of simultaneous Stability and Equity)**.** *Under Theorem 7, let* $G_\ell = \{j_1, \ldots, j_m\}$ *be a group of* $m \geq 2$ *features with pairwise correlation* ρ *and equal true coefficients. For any sequential gradient-boosted ensemble* f_ω *with* T *trees and any* $\rho \in (0, 1)$*:*

(i) γ*-Equity requires* $\gamma \to \infty$ *as* $\rho \to 1$*, i.e., the attribution ratio within* G_ℓ *diverges; or*

(ii) if Equity is enforced (e.g., by post-hoc normalization), then δ*-Stability requires* $\delta \to 1$*, i.e., rankings become maximally unstable.*

No choice of T, learning rate η, or tree structure eliminates this trade-off.

Proof. We prove both parts separately.

Part (i): Sequential boosting violates Equity.

Fix a seed ω. By symmetry of the DGP (Theorem 1), every feature in G_ℓ has the same marginal distribution and the same true coefficient. However, the greedy split selection in h_1 must break this symmetry: exactly one feature $j^*(\omega) \in G_\ell$ is selected at the root of h_1, determined by the random seed ω through sub-sampling and numerical tie-breaking.

By Theorem 6, this first-mover $j^*(\omega)$ accumulates a split advantage of $\Omega(\rho^2 \cdot T)$ over every other member of G_ℓ. By Theorem 9 and the limits (10), as $\rho \to 1$:

$$\phi_{j_q}(f_\omega) = \Theta\left(\frac{1-\rho^2}{2-\rho^2}\right) \xrightarrow{\rho\to 1} 0, \tag{13}$$

while $\phi_{j^*(\omega)}(f_\omega) = \Theta(1/(2-\rho^2)) \to \Theta(1)$. Therefore the ratio diverges:

$$\frac{\phi_{j^*(\omega)}(f_\omega)}{\phi_{j_q}(f_\omega)} = \Theta\left(\frac{1}{1-\rho^2}\right) \xrightarrow{\rho\to 1} \infty, \tag{14}$$

establishing genuine divergence ($\gamma \to \infty$). Hence no single model f_ω is γ-equitable for bounded γ as $\rho \to 1$.

Part (ii): Equity enforcement destroys Stability.

Suppose one attempts to restore equity by some mechanism that produces $\phi_j \approx \phi_k$ for all $j, k \in G_\ell$ (e.g., permutation-based renormalization, or averaging within the group for a single model). Consider what this implies for the ranking of features across G_ℓ.

For two independent seeds ω and ω', the first-movers $j^*(\omega)$ and $j^*(\omega')$ are drawn approximately uniformly from G_ℓ (by DGP symmetry). With probability $1 - 1/m$, they differ: $j^*(\omega) \neq j^*(\omega')$.

Without equity enforcement, the importance ranking within G_ℓ is determined by the first-mover identity. When $j^*(\omega) \neq j^*(\omega')$, the top-ranked feature within G_ℓ differs between the two runs. A rank displacement of magnitude $\Theta(m)$ for the first-mover feature, combined with the resulting within-group reshuffling of m features, contributes $\Theta(m^3)$ to $\sum_i d_i^2$.[1] By the Spearman formula $(1 - 6\sum_i d_i^2/(P(P^2-1)))$:

$$\text{Spearman}\big(\boldsymbol{\phi}(f_\omega),\ \boldsymbol{\phi}(f_{\omega'})\big) \ \le\ 1 - \Omega\left(\frac{m^3}{P^3}\right) \ =\ 1 - \Omega\left(\left(\frac{m}{P}\right)^3\right). \tag{15}$$

For balanced groups ($m = P/L$, L fixed), this is $1 - \Omega(L^{-3})$, a constant strictly less than 1.

With equity enforcement, the attribution gap within G_ℓ is compressed to near zero. But the rank *ordering* of near-tied features is then determined by residual noise of order $O(1/\sqrt{n})$ in the SHAP estimates, which is independent across seeds. The within-group ranking therefore becomes effectively random:

$$\mathbb{E}\big[\text{Spearman}_{G_\ell}\big(\boldsymbol{\phi}(f_\omega),\ \boldsymbol{\phi}(f_{\omega'})\big)\big] \ \to\ 0 \quad \text{as } \rho \to 1. \tag{16}$$

In either case, stability degrades with ρ.

Combining both parts.

A single sequential gradient-boosted model f_ω faces a dichotomy:

[1] For a group of m features whose ranks are permuted within a range of width m, a complete within-group rank reversal changes $\sum_i d_i^2$ by $\Theta(m^3)$.

- *Accept the natural split pattern*: Equity is violated by $\Omega(\rho^2)$ (Theorems 6 and 9).
- *Enforce equitable attributions*: The resulting near-ties make within-group rankings noise-dominated, and since the noise is independent across seeds, Stability is violated.

No finite T, learning rate η, or tree depth d can resolve this, because the first-mover advantage in Theorem 6 holds for any $T \geq 1$ and any $\eta > 0$, and the symmetry-breaking in h_1 is an inherent consequence of greedy sequential optimization. □

5 DASH Circumvention

Corollary 11 (DASH achieves equity and between-group stability)**.** *Let $f_{\omega_1}, \ldots, f_{\omega_M}$ be M independently trained gradient-boosted models with i.i.d. seeds $\omega_1, \ldots, \omega_M$. Define the* DASH *consensus attribution as*

$$\bar{\phi}_j = \frac{1}{M} \sum_{i=1}^{M} \phi_j(f_{\omega_i}). \tag{17}$$

Define also the group-level consensus importance $\bar{\Phi}_\ell = |G_\ell|^{-1} \sum_{j \in G_\ell} \bar{\phi}_j$. Then:

(a) ***Equity in expectation.*** *By symmetry of the DGP and independence of the seeds, for any $j, k \in G_\ell$,*

$$\mathbb{E}[\bar{\phi}_j] = \mathbb{E}[\bar{\phi}_k]. \tag{18}$$

The first-mover identity $j^*(\omega_i)$ is approximately uniformly distributed over G_ℓ, so each feature serves as first-mover in approximately M/m of the M models. The attribution concentration from Theorem 9 cancels in expectation.

(b) ***Between-group stability via the law of large numbers.*** *Define the consensus group-level importance $\bar{\Phi}_\ell = \frac{1}{|G_\ell|} \sum_{j \in G_\ell} \bar{\phi}_j$. By (a), $\mathbb{E}[\bar{\phi}_j]$ is the same for all $j \in G_\ell$, so $\mathbb{E}[\bar{\Phi}_\ell] = \mathbb{E}[\phi_j(f_\omega)]$ for any $j \in G_\ell$. For groups with distinct $\beta_\ell \neq \beta_{\ell'}$, these limits are distinct. The variance satisfies*

$$\mathrm{Var}(\bar{\Phi}_\ell) \leq \frac{1}{M|G_\ell|} \mathrm{Var}\big(\phi_j(f_{\omega_1})\big) = O(1/M). \tag{19}$$

By the law of large numbers, $\bar{\Phi}_\ell \xrightarrow{\text{a.s.}} \mathbb{E}[\Phi_\ell]$ for each ℓ, and two independent DASH *runs yield $\bar{\mathbf{\Phi}} = (\bar{\Phi}_1, \ldots, \bar{\Phi}_L)$ and $\bar{\mathbf{\Phi}}'$ satisfying*

$$\mathbb{E}\big[\mathrm{Spearman}\big(\bar{\mathbf{\Phi}},\ \bar{\mathbf{\Phi}}'\big)\big] \ \to\ 1 \quad \text{as } M \to \infty. \tag{20}$$

(c) ***Within-group instability is irreducible.*** *For $j, k \in G_\ell$ with $\beta_j = \beta_k$, the consensus difference $\bar{\phi}_j - \bar{\phi}_k$ has mean 0 and variance $\Theta(1/M)$ that does not vanish; each of two independent runs ranks j above k with probability $\to 1/2$ regardless of M. This is not a limitation of* DASH *but an irreducible consequence of the underlying symmetry: when the DGP assigns equal importance to j and k, no attribution method can consistently distinguish them.* DASH *makes this symmetry explicit (via equity) rather than masking it with an arbitrary first-mover ranking.*

DASH *thus achieves equity and between-group stability simultaneously, resolving the aspect of the impossibility that matters for the interpretability goal, by replacing a single sequential model with an average over an independent ensemble and thereby breaking the conditions of Theorem 10.*

Proof. Part (a) follows from linearity of expectation and the symmetry of the DGP under permutation of features within each group G_ℓ. Since the seeds ω_i are i.i.d., the probability that $j^*(\omega_i) = j$ is $1/m$ for each $j \in G_\ell$ and each i. Hence $\mathbb{E}[\phi_j(f_{\omega_i})] = \mathbb{E}[\phi_k(f_{\omega_i})]$ for all $j, k \in G_\ell$, and linearity gives $\mathbb{E}[\bar{\phi}_j] = \mathbb{E}[\bar{\phi}_k]$.

Part (b) follows from the law of large numbers applied to the group-level importances $\bar{\Phi}_\ell$. Since all features in G_ℓ share the same expected importance by (a), $\mathbb{E}[\bar{\Phi}_\ell] = \mathbb{E}[\phi_j(f_\omega)]$ for any $j \in G_\ell$, and this value increases with β_ℓ (groups with larger coefficients attract more splits by the gain criterion). The variance bound $\mathrm{Var}(\bar{\Phi}_\ell) \leq 1/(M|G_\ell|) \cdot \mathrm{Var}(\phi_j(f_{\omega_1}))$ follows from i.i.d. seeds and the bound $\mathrm{Var}(\bar{\Phi}_\ell) = \mathrm{Var}\big(\frac{1}{M|G_\ell|} \sum_{i,j} \phi_j(f_{\omega_i})\big) \leq \frac{1}{M|G_\ell|}\mathrm{Var}(\phi_j)$ using i.i.d. seeds and the covariance inequality (features within the same group in the same model are positively correlated, but the bound uses only the marginal variance, giving a conservative upper bound). Concentration in probability gives convergence to distinct limits across groups, ensuring the group-level rank order stabilises.

Part (c) follows from the symmetry of (a): $\bar{\phi}_j - \bar{\phi}_k \sim N(0, \Theta(1/M))$ by CLT (independent seeds, bounded variance), and two independent runs produce independent draws of this quantity; the probability they share the same sign is $1/2 + O(1/\sqrt{M})$, not tending to 1. □

Remark 12 (Within-group instability is not a pathology). Theorem 11(c) may appear to concede ground—DASH achieves equity but leaves within-group rankings noise-dominated. However, this is the *correct* behaviour when features are genuinely equivalent in the DGP: any attribution method that consistently ranks j above k when $\beta_j = \beta_k$ is introducing a spurious asymmetry. The relevant interpretability target is identifying *which group* is important (between-group stability) and confirming that no individual feature within the group is systematically advantaged (equity). Both are achieved. Individual within-group rankings are interpretively meaningless when $\beta_j = \beta_k$, and the appropriate diagnostic for detecting within-group instability is the FSI (Feature Stability Index, defined in the main paper), not rank correlation.

6 Discussion of Assumptions

Remark 13 (Sequential structure is essential). The impossibility in Theorem 10 relies critically on the *sequential* nature of gradient boosting, where each tree h_t is fit to residuals $r_t = y - \sum_{s<t} h_s(x)$. This creates a causal chain: the first tree's split choices alter the residual landscape for all subsequent trees, producing the cumulative first-mover advantage of Theorem 6.

The theorem does *not* apply to methods that train base learners independently. For example:

- **Random forests** train each tree on a bootstrap sample with independent feature subsampling. Since trees do not share residuals, there is no first-mover propagation: if tree t splits on j, this does not affect subsequent trees' feature selection. The equity violation is therefore $O(1)$ rather than $O(\rho^2 \cdot T)$.

- **Bagged ensembles** similarly avoid the sequential dependency by construction.

However, gradient boosting's sequential structure is precisely what makes it a powerful function approximator (by fitting residuals, it can represent complex interactions that parallel ensembles miss). The impossibility result thus highlights a fundamental tension between the *modeling power* of sequential boosting and the *fairness properties* of feature attribution.

Remark 14 (Scope of Theorem 7). Theorem 7 (the uniform-contribution model) abstracts away the depth-weighting of TreeSHAP, which assigns higher weight to splits closer to the root. In practice, early splits on the first-mover feature $j^*(\omega)$ occur preferentially near the root (since the first tree's

root split is on $j^*(\omega)$), so the actual attribution gap is likely *larger* than the $\Omega(\rho^2)$ established here. The assumption therefore gives a conservative lower bound. A tighter analysis tracking depth-weighted split counts would strengthen the impossibility result.